

Chapter 3

Handout 3

SUPPLY AND DEMAND: THEORY

INTRODUCTION

Supply and demand have been called the “bread and butter” of economics. A market is any place people come together to trade. Economists often say that every market has two sides: a buying side and a selling side. The buying side of the market is usually referred to as the demand side; the selling side is usually referred to as the supply side.

Demand refers to the willingness and ability of buyers to purchase different quantities of a good at different prices during a specific period (per day, week, etc.).

It implies that to be demand consumer must have **(1) Desire (2) Plan and (3) Ability**.

The Law of Demand By four methods we can define the law of demand: **(1) Descriptive** method **(2) Symbolic** method) **(3) Numerical** method or by schedule **(4) Graphical** method.

- (1) Descriptive method:** The law of demand states that as the price of a good rises, the quantity demanded of the good falls, and that as the price of a good falls, the quantity demanded of the good rises, *ceteris paribus* or other things remaining same. That means, the price of a good and the quantity demanded of it are inversely related, *ceteris paribus*.

- (2) Symbolic Method**

$$P \uparrow Q_d \downarrow$$

$$P \downarrow Q_d \uparrow, \text{ceteris paribus}$$

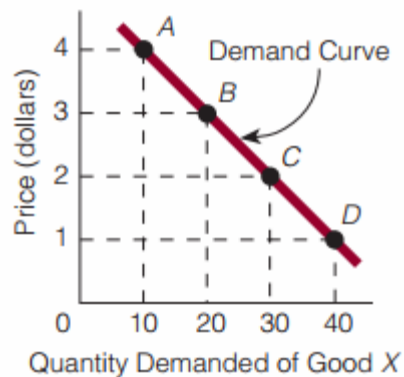
where P = price and Q_d = quantity demanded.

- (3) Numerical Method**

Demand Schedule for Good X		
Price (dollars)	Quantity Demanded	Point in Part (b)
4	10	A
3	20	B
2	30	C
1	40	D

A demand schedule is the numerical representation of the law of demand.

- (4) Graphical Method**



The four price–quantity combinations are plotted and the points connected, giving us a (downward-sloping) demand curve. A (downward-sloping) demand curve is the graphical representation of the inverse relationship between price and quantity demanded specified by the law of demand. In short, a demand curve is a picture of the law of demand.

Why quantity demanded moves in the direction opposite that of price or why demand curve is downward sloping?

We identify two reasons:

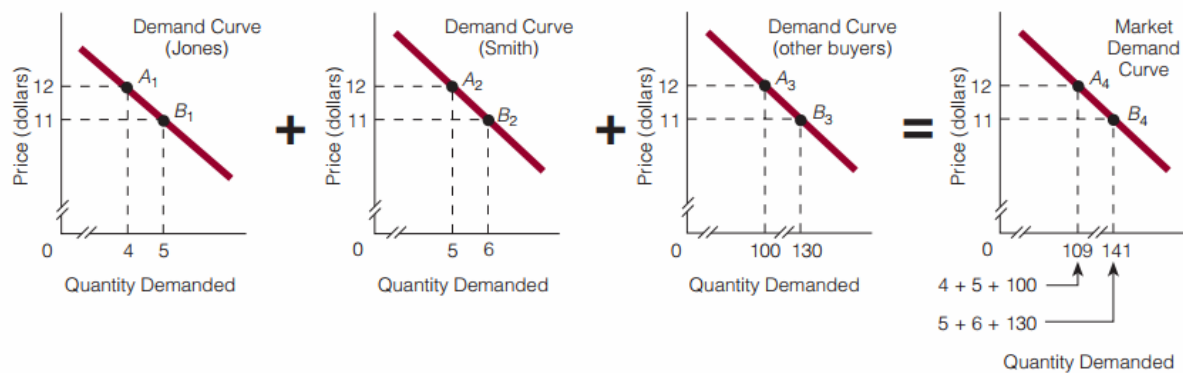
- (1) People substitute lower priced goods for higher priced goods. Suppose orange juice and grapefruit juice are substitutes. If price of orange juice rises, people will choose to buy less of the relatively higher priced orange juice and more of the relatively lower priced grapefruit juice. In other words, a rise in the price of orange juice will lead to a decrease in the quantity demanded of it.
- (2) The second reason for the inverse relationship between price and quantity demanded has to do with the law of diminishing marginal utility, which states that, over a given period, the marginal (or additional) utility or satisfaction gained by consuming equal successive units of a good will decline as the amount of the good consumed increases.

What does marginal utility have to do with the law of demand? Economists state that, the more utility you receive from a unit of a good, the higher the price you are willing to pay for it, and the less utility you receive from a unit of a good, the lower the price you are willing to pay for it. According to the law of diminishing marginal utility, individuals obtain less utility from additional units of a good. Therefore, they will buy larger quantities of a good only at lower prices, and their behavior reflects the law of demand.

Individual Demand Curve and Market Demand Curve An individual demand curve represents the price–quantity combinations of a particular good for a single buyer. By contrast, a market demand curve represents the price–quantity combinations of a good for all buyers. A market demand curve is derived by adding up individual demand curves. The demand schedules for Jones, Smith, and other buyers are shown in the following table. The market demand schedule is obtained by adding the quantities demanded at each price. For example, at \$12, the quantities demanded are 4 units for Jones, 5 units for Smith, and 100 units for other buyers. Thus, a total of 109 units are demanded at \$12.

Price	Quantity Demanded						
	Jones		Smith		Other Buyers		All Buyers
\$15	1		2		20		23
14	2		3		45		50
13	3		4		70		77
12	4	+	5	+	100	=	109
11	5	+	6	+	130	=	141
10	6		7		160		173

The data points for the demand schedules are plotted and added to produce a market demand curve. The market demand curve could also be drawn directly from the market demand schedule.



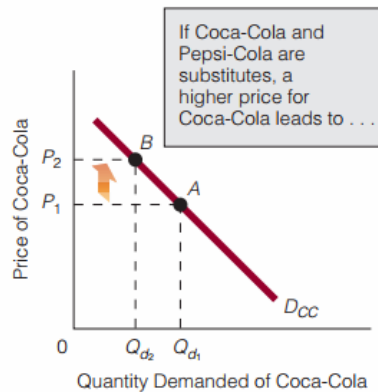
Major Determinants of Demand

- (1) Price of the commodity
- (2) Prices of related goods
- (3) Income of the consumer
- (4) Taste and preferences of the consumer
- (5) Expected future price of the commodity.

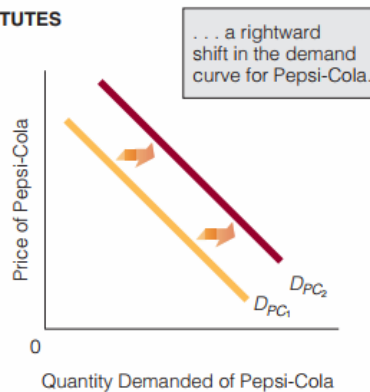
- (1) **Price of the commodity:** as the price of a good rises, the quantity demanded of the good falls, and that as the price of a good falls, the quantity demanded of the good rises, ceteris paribus or other things remaining same. That means, the price of a good and the quantity demanded of it are inversely related, ceteris paribus.
- (2) **Prices of Related Goods:** There are two types of related goods: substitutes and complements. Two goods are substitutes if they satisfy similar needs or desires. For many people, Coca-Cola and Pepsi-Cola are substitutes. If two goods are substitutes, then, as the price of one rises (falls), the demand for the other rises (falls). For instance, higher Coca-Cola prices will increase the demand for Pepsi-Cola as people substitute Pepsi for the higher priced Coke.

Other examples of substitutes are coffee and tea, corn chips and potato chips, different brands of margarine, and foreign and domestic cars. Generalizing, we obtain

X and Y are substitutes: If $P_X \uparrow$, then $D_Y \uparrow$ If $P_X \downarrow$, then $D_Y \downarrow$



SUBSTITUTES



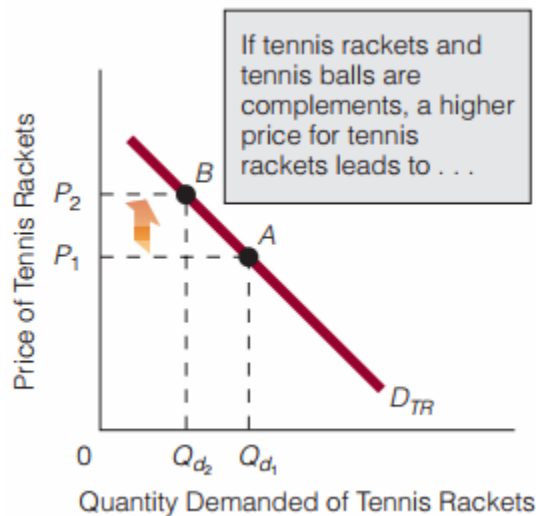
Substitutes and Complements

(a) Coca-Cola and Pepsi-Cola are substitutes: The price of one and the demand for the other are directly related. As the price of Coca-Cola rises, the demand for Pepsi-Cola increases. (b) Tennis rackets and tennis balls are complements: The price of one and the demand for the other are inversely related. As the price of tennis rackets rises, the demand for tennis balls decreases.

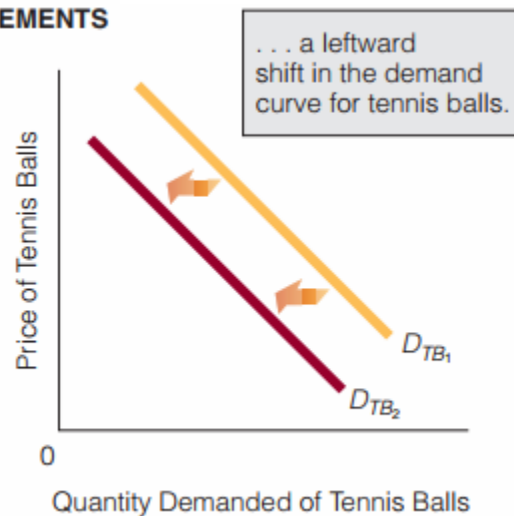
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Two goods are complements if they are consumed jointly. For example, tennis rackets and tennis balls are used together to play tennis. If two goods are complements, then, as the price of one rises (falls), the demand for the other falls (rises). For example, higher tennis racket prices will decrease the demand for tennis balls. Other examples of complements are cars and tires left shoe and right shoe. Generalizing yields the following relationship:

X and Y are complements: If $P_X \uparrow$, then $D_Y \downarrow$ If $P_X \downarrow$, then $D_Y \uparrow$



COMPLEMENTS



- (3) Income:** As a person's income changes (increases or decreases), that individual's demand for a particular good may rise, fall, or remain constant. For example, suppose Jack's income rises and, as a consequence, his demand for CDs rises. Then, for Jack, CDs are a normal good. For a normal good, demand rises as income rises and demand falls as income falls:

X is a normal good: If income $\uparrow$, then $D_X \uparrow$ If income $\downarrow$, then $D_X \downarrow$

Now suppose Marie's income rises and, as a consequence, her demand for canned baked beans falls. Then, for Marie, canned baked beans are an inferior good. For an inferior good, demand falls as income rises and demand rises as income falls:

Y is an inferior good: If income $\uparrow$, then $D_Y \downarrow$ If income $\downarrow$, then $D_Y \uparrow$

Finally, suppose that, when George's income rises, his demand for toothpaste neither rises nor falls. Then, for George, toothpaste is neither a normal good nor an inferior good. Instead, it is a neutral good. For a neutral good, demand does not change as income rises or falls.

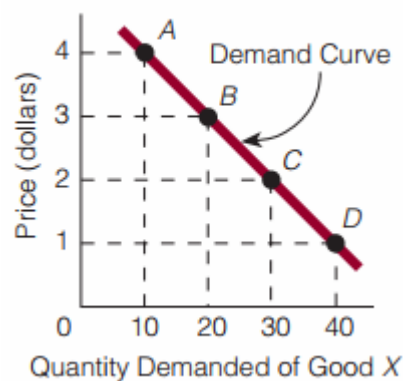
(4) Preferences: People's preferences affect the amount of a good they are willing to buy at a particular price. A change in preferences in favor of a good shifts the demand curve rightward. A change in preferences away from the good shifts the demand curve leftward. For example, if people begin to favor Elmore Leonard novels to a greater degree than previously, the demand for his novels increases and the demand curve shifts rightward.

(5) Expectations of Future Price: Buyers who expect the price of a good to be higher next month may buy it now, thus increasing the current (or present) demand for the good. Buyers who expect the price of a good to be lower next month may wait until next month to buy it, thus decreasing the current (or present) demand for the good. For example, suppose you are planning to buy a house. One day, you hear that house prices are expected to go down in a few months. Consequently, you decide to delay your purchase for a while. Alternatively, if you hear that prices are expected to rise in a few months, you might go ahead and make your purchase now.

A Change in Quantity Demanded Versus a Change in or Shift in Demand

Quantity demanded = Number of units of a good that individuals are willing and able to buy at a particular price

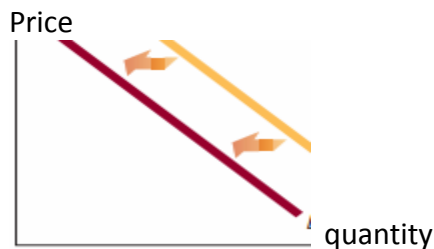
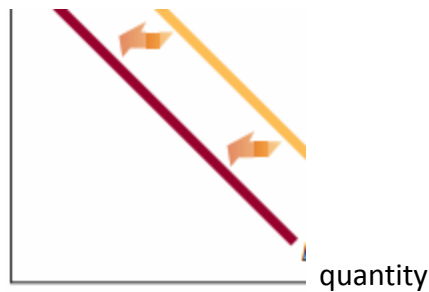
Change in quantity demanded = A movement from one point to another point on the same demand curve that is caused by a change in the own price of the good. That means movement along the same demand due to change in price but other things remain constant is known as



change in quantity demanded.

In the above figure, movement from A to D or from D to A along the same demand curve due to change in price of good X is known as change in quantity demanded.

Price



A change or shift in demand implies parallel rightward or leftward shift of entire demand curve. Change in (1) income (2) preferences (3) prices of related goods (4) the number of buyers and (5) expectations of future prices cause shift in demand.

For example, if income rises demand curve shifts to the right and if income falls demand curve shifts to the left. If preferences go up demand curve shifts to the right and If preferences goes down demand curve shifts to the left.

Policy makers use these two concepts to reduce smoking. Policy makers use tax and advertisement as instruments. By imposing higher tax on cigarettes they raise price which reduces demand for cigarette. It implies movement along demand curve known as change in quantity demanded. When policy makers advertise by saying that 'smoking causes lung cancer' people' preference and taste change, people less amount of cigarette and demand curve shift to the left.

Supply

The willingness and ability of sellers to produce and offer to sell different quantities of a good at different prices during a specific period is known as supply.

The Law of Supply

By four methods we can define the law of supply

1. **Descriptive method:** As the price of a good rises, the quantity supplied of the good rises, and as the price of a good falls, the quantity supplied of the good falls, *ceteris paribus* i.e., the price of a good and the quantity supplied of the good are directly related, *ceteris paribus*. It implies that the supply curve is upward sloping.
2. **Symbolical Method**

$$P \uparrow Q_s \uparrow$$

$$P \downarrow Q_s \downarrow, \text{ ceteris paribus}$$

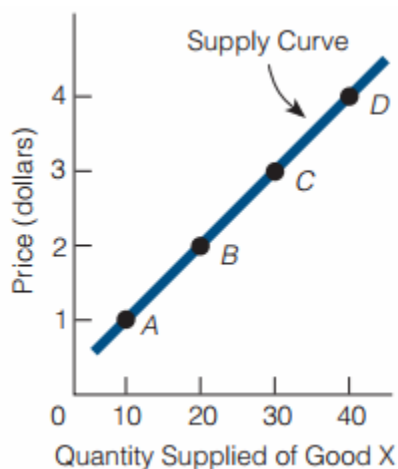
where P = price and Q_s = quantity supplied.

3. Numerical Method or by schedule

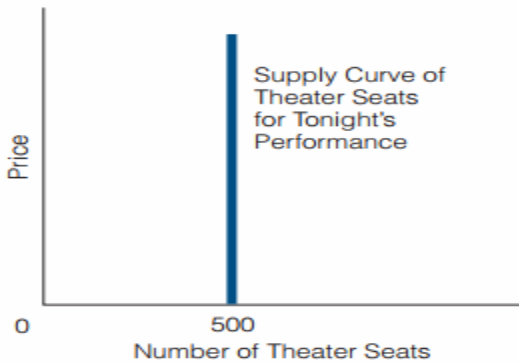
P	Q_s
1	10
2	20
3	30
4	40

In the above table we see that as the price of a good rises, the quantity supplied of the good rises, and as the price of a good falls, the quantity supplied of the good falls.

4. Graphical Method



The upward-sloping supply curve is the graphical representation of the law of supply, which states that price and quantity supplied are directly related, *ceteris paribus*. On a supply curve, the price (in dollars) represents the price per unit of the good. The quantity supplied, on the horizontal axis, is always relevant for a specific period (a week, a month, etc.).



The supply curve is not upward sloping when there is no time to produce additional units or when additional units cannot be produced. In those cases, the supply curve is vertical.

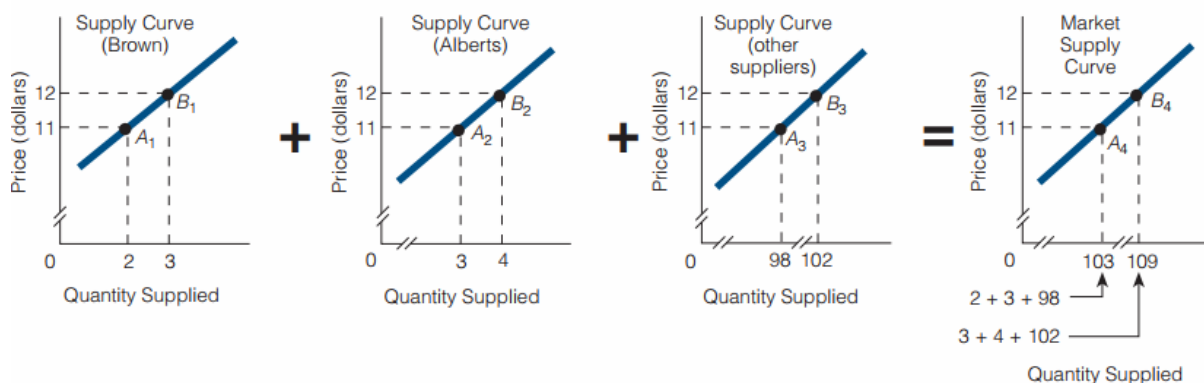
Why is a supply curve upward sloping?

Here, it to say that an upward-sloping supply curve reflects the fact that, under certain conditions, a higher price is an incentive to producers more of the good. But actual reason is the law diminishing marginal return.

The Market Supply Curve: An individual supply curve represents the price–quantity combinations for a single seller. The market supply curve represents the price–quantity combinations for all sellers of a particular good. Exhibit 8 shows how a market supply curve can be derived by adding individual supply curves. In part (a), a supply schedule, the numerical tabulation of the quantity supplied of a good at different prices is given for Brown, Alberts, and other suppliers. The market supply schedule is obtained by adding the quantities supplied at each price, *ceteris paribus*. For example, at \$11, the quantities supplied are 2 units for Brown, 3 units for Alberts, and 98 units for the other suppliers. Thus, 103 units are supplied at \$11. In part (b), the data points for the supply schedules are plotted and added to produce a market supply curve (which could also be drawn directly from the market supply schedule).

Price	Quantity Supplied					
	Brown		Alberts		Other Suppliers	All Suppliers
11	2	+	3	+	98	= 103
12	3	+	4	+	102	= 109

(a)



(b)

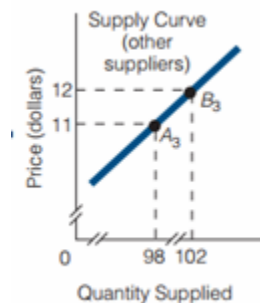
Determinants of Supply

1. Price of the commodity
2. Prices of resources
3. Technology
4. Number of sellers
5. Expected future price of the commodity.

We already learned direct relationship between price of the commodity and quantity supplied, *ceteris paribus*.

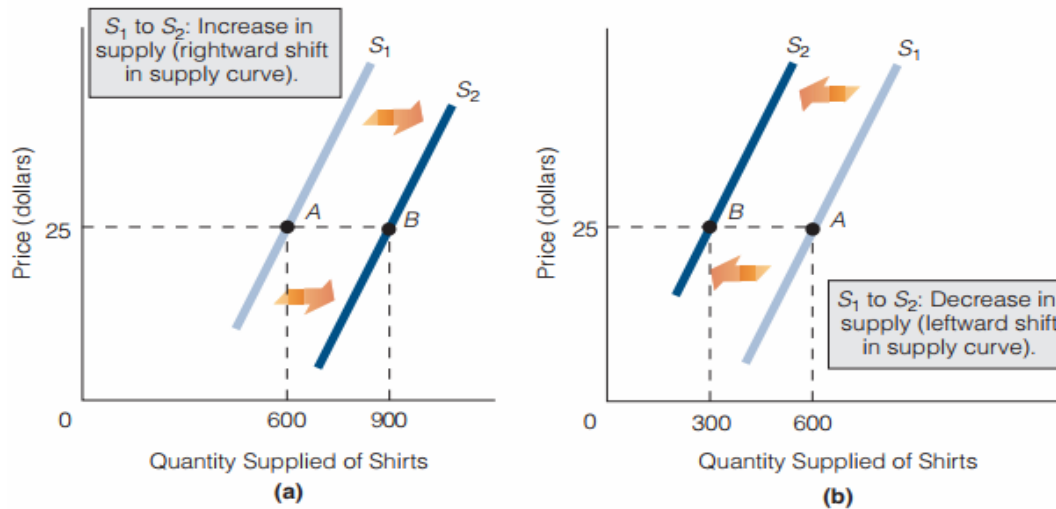
Change in quantity supplied vs change in or shift in supply

If price of the commodity changes when other things remain same, a seller move along the same supply curve as shown in following figure is known as **change in quantity supplied**.

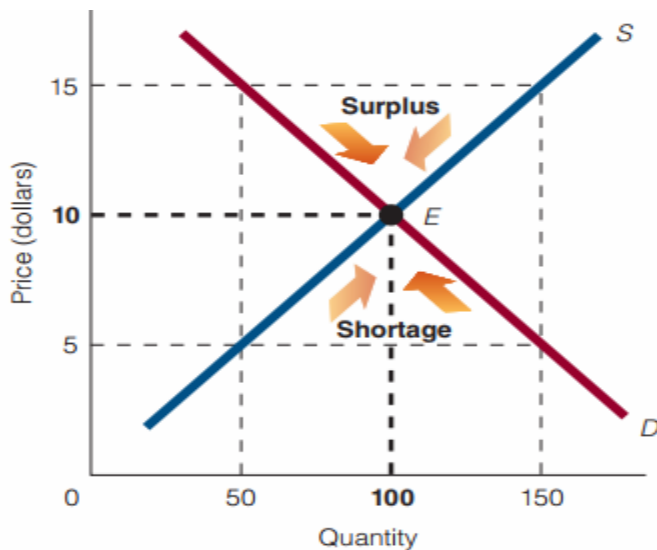


Change in supply or shift in supply

If other things such as prices of resources, technology, number of sellers and expected future price change, supply curve parallelly shift leftward or rightward. If prices of resources rise cost of production increases, seller produce less of the commodity and supply curve parallelly shift to the left and vice versa as shown in the following figures. If technology improves more output can be produced from the same inputs, supply curve shift to the right. If number of sellers increase supply increases and vice versa, supply curve shift to the right or to the left.



Market Equilibrium



When two opposite forces are in balance is known as equilibrium. Demand and supply are two opposite forces of a market. They met at E. E is equilibrium point where demand = Supply. Equilibrium price is 10 and equilibrium quantity is 100. Equilibrium price 10 is known as market clearing price since at this price there exists no surplus or shortage in the market. If price is more or less than equilibrium price disequilibrium arises in the market. If price is 15 surplus or

excess supply occurs and at price 5 shortage or excess demand occurs. If surplus arises seller will reduce price to sell excess quantity of output. Seller will continue to reduce until market comes back to equilibrium. In case of excess demand price will continue to rise until market comes back to equilibrium.. This mechanism is known as adjustment process.

What Can Change Equilibrium Price and Quantity?

Equilibrium price and quantity are determined by supply and demand. Whenever demand changes, or supply changes, or both change, equilibrium price and quantity change. Exhibit 18 illustrates eight different cases where this scenario occurs. Cases (a)–(d) illustrate the four basic changes in supply and demand: either supply changes or demand changes. Cases (e)–(h) illustrate changes in both supply and demand.

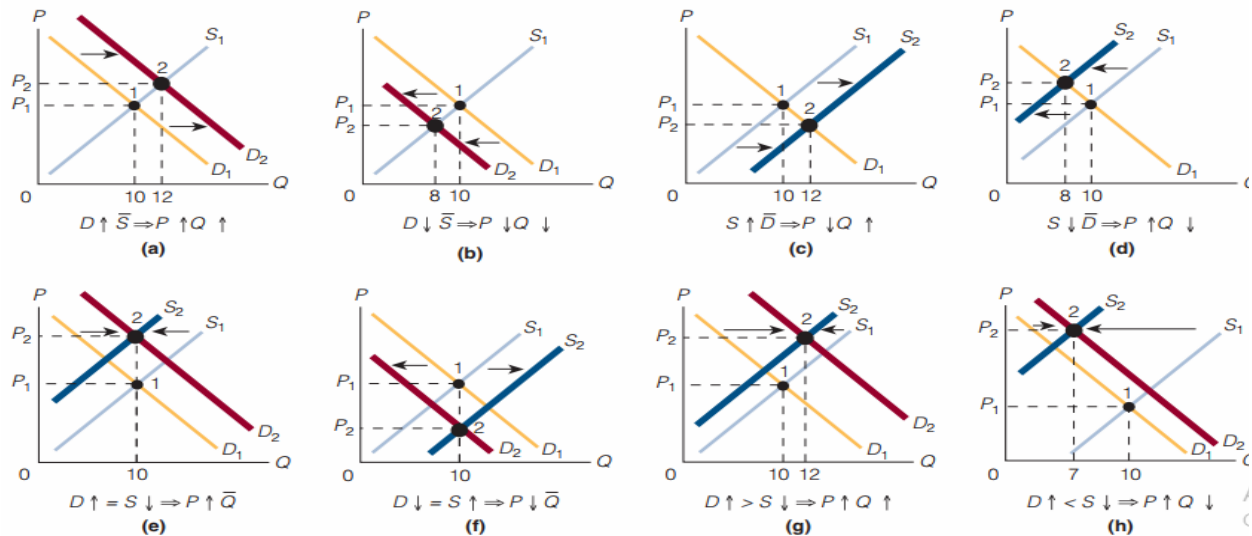
- Case (a): Demand rises (the demand curve shifts rightward from D_1 to D_2 , and supply is constant (the supply curve does not move). As a result of demand rising and supply remaining constant, the equilibrium price rises from P_1 to P_2 and the equilibrium quantity rises from 10 units to 12 units. Now, let's see if you can identify what has happened to quantity supplied (not supply) as price has risen from P_1 to P_2 . (Remember, quantity supplied changes if price changes.) As price rises from P_1 to P_2 , quantity supplied rises from 10 units to 12 units. We see this situation as a movement up the supply curve from point 1 to point 2, corresponding (on the horizontal axis) to a change from 10 units to 12 units.

Equilibrium Price and Quantity Effects of Supply Curve Shifts and Demand Curve Shifts

The exhibit illustrates the effects on equilibrium price and quantity of a change in

demand, a change in supply, and a change in both. Below each diagram, the condition leading to the effects is noted, using the following symbols: (1) a bar over a letter means constant (thus, $\bar{S}$ means that supply is constant); (2) a downward-pointing arrow ($\downarrow$)

indicates a fall; and (3) an upward-pointing arrow ($\uparrow$) indicates a rise. A rise (fall) in demand is the same as a rightward (leftward) shift in the demand curve. A rise (fall) in supply is the same as a rightward (leftward) shift in the supply curve.



DEMAND AND SUPPLY AS EQUATIONS

You are used to seeing demand and supply as curves. Let's now look at demand and supply as equations. Here is a demand equation:

$$Q_d = 1,500 - 32P$$

To see what this equation says, we let price (P) in the equation equal \$10 and then solve for quantity demanded (Q_d). We get 1,180.

$$Q_d = 1,500 - 32(10) = 1,180$$

So this equation says that if price is \$10, it follows that quantity demanded is 1,180 units. We could find other quantities demanded by plugging in different dollar amounts for price (P).

Now here is a supply equation:

$$Q_s = 1,200 + 43P$$

To find what quantity supplied (Q_s) equals at a particular price, we let \$5 equal price (P) and solve for quantity supplied. We get 1,415.

$$Q_s = 1,200 + 43(5) = 1,415$$

Now suppose we want to find equilibrium price and quantity given our demand and supply equations. How would we do it?

First, we know that in equilibrium the quantity demanded (Q_d) of a good is equal to the quantity supplied (Q_s), so let's set the two equations equal to each other this way:

$$1,500 - 32P = 1,200 + 43P$$

Now we can solve for P . We add $32P$ to both sides of the equal sign and subtract 1,200 from both sides. We are left with:

$$75P = 300$$

It follows then that $P = 300/75$ or \$4.00.

Once we know equilibrium price is \$4.00, we can place this value in either the demand or supply equation to find the equilibrium quantity. Let's place it in the demand equation:

$$Q_d = 1,500 - 32(4.00) = 1,372$$

Just to make sure that 1,372 is also the quantity supplied, we put the equilibrium price of \$4.00 into the supply equation:

$$Q_s = 1,200 + 43(4.00) = 1,372$$

In summary, given our demand and supply equations, equilibrium price is \$4.00 and equilibrium quantity is 1,372.

Sample questions for quiz1 from chapter-3

(1) What are the two sides of a market?

Buying or demand side and selling or supply side.

(2) What do you mean by demand?

Demand refers to the willingness and ability of buyers to purchase different quantities of a good at different prices during a specific period (per day, week, etc.).

(3) What are the conditions to be a demand? Consumer must have (1) desire (2) Plan and (3) Ability.

(4) Write down the law of demand.

The law of demand states that as the price of a good rises, the quantity demanded of the good falls, and that as the price of a good falls, the quantity demanded of the good rises, ceteris paribus

(5) What are the methods we use to explain the law of demand?

(1) Descriptive method (2) Symbolic method) (3) Numerical method or by schedule (4) Graphical method.

(6) Why demand curve is downward sloping? (a) People substitute lower priced goods for higher priced goods. **(b)** The law of diminishing marginal utility.

(7) What do you mean by the law of diminishing marginal utility?

According to the law of diminishing marginal utility, individuals obtain less utility from additional units of a good.

(8) What do you mean by a market demand curve?

A market demand curve represents the price–quantity combinations of a good for all buyers. A market demand curve is derived by adding up individual demand curves.

(9) What do you mean by change in quantity demanded?

A movement from one point to another point on the same demand curve that is caused by a change in the own price of the good is as change in quantity demanded.

(10) What do you mean by change in or shift in demand?

A change or shift in demand implies parallel rightward or leftward shift of entire demand curve due to change in other things such as (a) income (b) preferences (c) prices of related goods (d) expectations of future prices.

(11) What do you mean by the law of supply?

As the price of a good rises, the quantity supplied of the good rises, and as the price of a good falls, the quantity supplied of the good falls, *ceteris paribus*.

(12) Why is a supply curve upward sloping?

- (a) A higher price is an incentive to producers more of the good. (b) The law diminishing marginal return.

(13) What is the condition of market equilibrium?

Demand = Supply

Sample questions for Mid1 from chapter-3

1. Describe the law of demand by all possible methods

By four methods we can define the law of demand: **(1)** Descriptive method **(2)** Symbolic method **(3)** Numerical method or by schedule **(4)** Graphical method.

- (1)** Descriptive method: The law of demand states that as the price of a good rises, the quantity demanded of the good falls, and that as the price of a good falls, the quantity demanded of the good rises, *ceteris paribus* or other things remaining same. That means, the price of a good and the quantity demanded of it are inversely related, *ceteris paribus*.

(2) Symbolic Method

$$P \uparrow Q_d \downarrow$$

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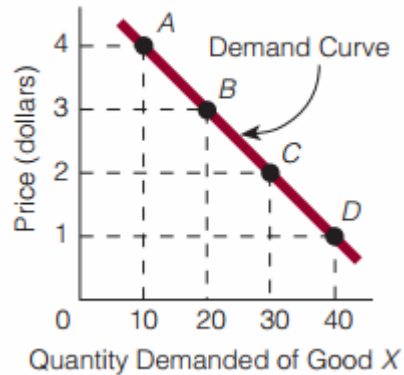
(3) Numerical Method

Demand Schedule for Good X

Price (dollars)	Quantity Demanded	Point in Part (b)
4	10	A
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A demand schedule is the numerical representation of the law of demand.

(4) Graphical Method



The four price–quantity combinations are plotted and the points connected, giving us a (downward-sloping) demand curve. A (downward-sloping) demand curve is the graphical representation of the inverse relationship between price and quantity demanded specified by the law of demand. In short, a demand curve is a picture of the law of demand.

2. Explain the major determinants of demand

Determinants of demand are (a) Price of the commodity (b) Prices of related goods (c) Income of the consumer (d) Taste and preferences of the consumer and (e) Expected future price of the commodity.

- (a) **Price of the commodity:** as the price of a good rises, the quantity demanded of the good falls, and that as the price of a good falls, the quantity demanded of the good rises, ceteris paribus or other things remaining same. That means, the price of a good and the quantity demanded of it are inversely related.
- (b) **Prices of Related Goods:** There are two types of related goods: substitutes and complements. Two goods are substitutes if they satisfy similar needs or desires. For many people, Coca-Cola and Pepsi-Cola are substitutes. If two goods are substitutes, then, as the price of one rises (falls), the demand for the other rises (falls). For instance, higher Coca-Cola prices will increase the demand for Pepsi-Cola as people substitute Pepsi for the higher priced Coke.

X and Y are substitutes: If $P_X \uparrow$, then $D_Y \uparrow$ If $P_X \downarrow$, then $D_Y \downarrow$

Two goods are complements if they are consumed jointly. For example, tennis rackets and tennis balls are used together to play tennis. If two goods are complements, then, as the price of one rises (falls), the demand for the other falls (rises). For example, higher tennis racket prices will decrease the demand for tennis balls.

X and Y are complements: If $P_X \uparrow$, then $D_Y \downarrow$ If $P_X \downarrow$, then $D_Y \uparrow$

- (c) **Income:** As a person's income changes (increases or decreases), that individual's demand for a particular good may rise, fall, or remain constant. For example, suppose Jack's income

rises and, as a consequence, his demand for CDs rises. Then, for Jack, CDs are a normal good. For a normal good, demand rises as income rises and demand falls as income falls:

X is a normal good: If income $\uparrow$, then $D_X \uparrow$ If income $\downarrow$, then $D_X \downarrow$

Now suppose Marie's income rises and, as a consequence, her demand for canned baked beans falls. Then, for Marie, canned baked beans are an inferior good. For an inferior good, demand falls as income rises and demand rises as income falls:

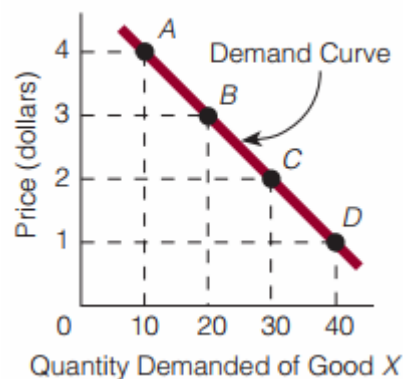
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- (d) Preferences:** People's preferences affect the amount of a good they are willing to buy at a particular price. A change in preferences in favor of a good shifts the demand curve rightward. A change in preferences away from the good shifts the demand curve leftward. For example, if people begin to favor Elmore Leonard novels to a greater degree than previously, the demand for his novels increases and the demand curve shifts rightward.
- (e) Expectations of Future Price:** Buyers who expect the price of a good to be higher next month may buy it now, thus increasing the current (or present) demand for the good. Buyers who expect the price of a good to be lower next month may wait until next month to buy it, thus decreasing the current (or present) demand for the good. For example, suppose you are planning to buy a house. One day, you hear that house prices are expected to go down in a few months. Consequently, you decide to delay your purchase for a while. Alternatively, if you hear that prices are expected to rise in a few months, you might go ahead and make your purchase now.

3. Distinguish between change in quantity demanded and change in or shift in demand. How do policy makers use these two concepts to reduce smoking?

Change in quantity demanded = A movement from one point to another point on the same demand curve that is caused by a change in the own price of the good. That means movement along the same demand due to change in price but other things remain constant is known as



change in quantity demanded.

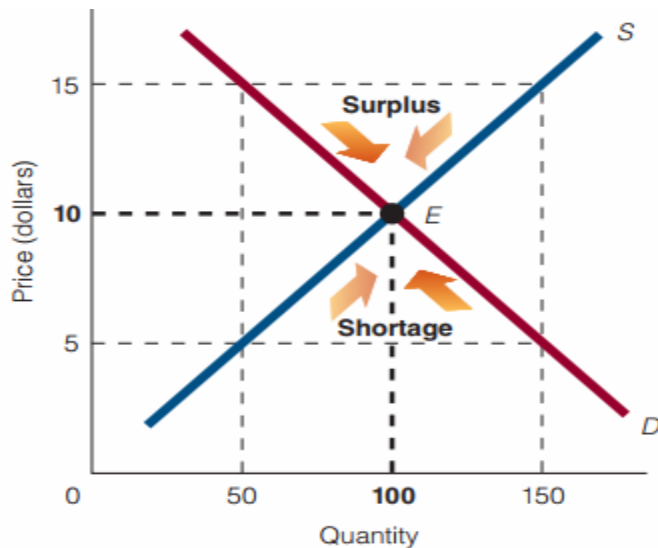
In the above figure, movement from A to D or from D to A along the same demand curve due to change in price of good X is known as change in quantity demanded.

A change or shift in demand implies parallel rightward or leftward shift of entire demand curve. Change in (1) income (2) preferences (3) prices of related goods (4) the number of buyers and (5) expectations of future prices cause shift in demand. For example, if income rises demand curve shifts to the right and if income falls demand curve shifts to the left. If preferences go up demand curve shifts to the right and If preferences goes down demand curve shifts to the left.



Policy makers use these two concepts to reduce smoking. Policy makers use tax and advertisement as instruments. By imposing higher tax on cigarettes they raise price which reduces demand for cigarette. It implies movement along demand curve known as change in quantity demanded. When policy makers advertise by saying that 'smoking causes lung cancer' people's preference and taste change, people consume less amount of cigarette and demand curve shifts to the left.

4. Explain the market equilibrium and the adjustment process if disequilibrium occurs.



When two opposite forces are in balance is known as equilibrium. Demand and supply are two opposite forces of a market. They met at E. E is equilibrium point where demand = Supply. Equilibrium price is 10 and equilibrium quantity is 100. Equilibrium price 10 is known as market clearing price since at this price there exists no surplus or shortage in the market. If price is more or less than equilibrium price disequilibrium arises in the market. If price is 15 surplus or excess supply occurs and at price 5 shortage or excess demand occurs. If surplus arises seller will reduce price to sell excess quantity of output. Seller will continue to reduce until market comes back to equilibrium. In case of excess demand price will continue to rise until market comes back to equilibrium.. This mechanism is known as adjustment process.

5. Determine equilibrium price and quantity from following demand and supply equations.

DEMAND AND SUPPLY AS EQUATIONS

You are used to seeing demand and supply as curves. Let's now look at demand and supply as equations. Here is a demand equation:

$$Q_d = 1,500 - 32P$$

To see what this equation says, we let price (P) in the equation equal \$10 and then solve for quantity demanded (Q_d). We get 1,180.

$$Q_d = 1,500 - 32(10) = 1,180$$

So this equation says that if price is \$10, it follows that quantity demanded is 1,180 units. We could find other quantities demanded by plugging in different dollar amounts for price (P).

Now here is a supply equation:

$$Q_s = 1,200 + 43P$$

To find what quantity supplied (Q_s) equals at a particular price, we let \$5 equal price (P) and solve for quantity supplied. We get 1,415.

$$Q_s = 1,200 + 43(5) = 1,415$$

Now suppose we want to find equilibrium price and quantity given our demand and supply equations. How would we do it?

First, we know that in equilibrium the quantity demanded (Q_d) of a good is equal to the quantity supplied (Q_s), so let's set the two equations equal to each other this way:

$$1,500 - 32P = 1,200 + 43P$$